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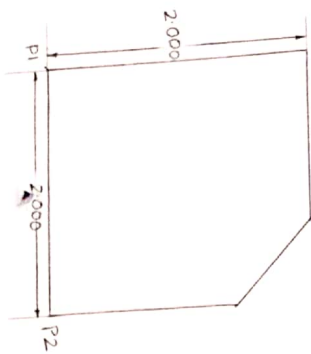
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DIMENSIONING

Linear dimensioning



EXPERIMENT NO. 3

DATE: 31.1.19

Roll No.: 1821052

TYPES OF DIMENSIONING

AIM: To draw and perform different types of dimensioning

THEORY:

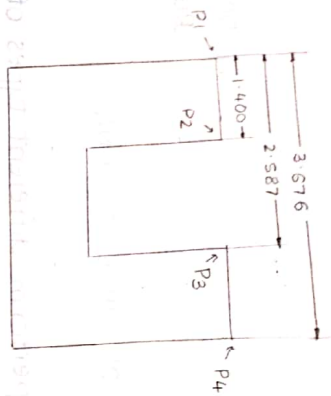
DIMENSIONING - A numerical value expressed in appropriate units of measurements and indicated graphically on technical drawings with lines, symbols and notes.

- Linear dimensioning
- Baseline dimensioning
- Continue dimensioning
- Diameter dimensioning
- Radius dimensioning
- Angular dimensioning
- Aligned dimensioning

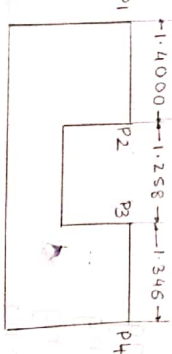
⇒ Linear dimensioning

- It allows you to create horizontal and vertical dimensions.
- Select the LINEAR command using the icon or dimension / linear command : dimlinear
- Specify first extension line origin or <select object>:
- Snap to first extension line origin P1
- Specify second extension line origin : Snap to P2
- Specify dimension line location : Select where you want dimension line placed P3
- The dimension test value will be displayed on the last line

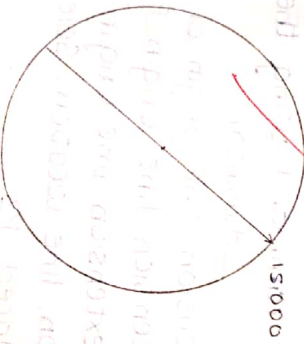
Baseline dimensioning



Continue dimensioning



Diameter dimensioning



⇒

Baseline dimensioning - It allows you to establish a baseline for successive dimensions. The spacing between dimensions is automatic and should be set in dimension style. A baseline dimension must be widely used with an existing dimension. If you use baseline dimensioning immediately after linear dimension, you do not have to specify baseline origin.

- Create a linear dimension first (1.400 P1 to P2)
- Select Baseline command using the icon or dimension/baseline command
- Specify a second extension line origin <select>: snap to P3
- Specify a second extension line origin <select>: snap to P4
- Select <center> twice to stop.

⇒

Continue dimensioning - It creates a series of dimension in line form and existing dimension. If you use continue dimensioning immediately after a linear dimension, you do not have to specify continue extension origin.

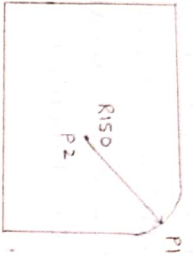
- Create a linear dimension first (1.400 P1 and P2)
- Select CONTINUE command using icon or dimension/continue command
- Specify a second extension line origin <select>: snap to P3
- Specify a second extension line origin <select>: snap to P4
- Press <center> twice to stop

⇒

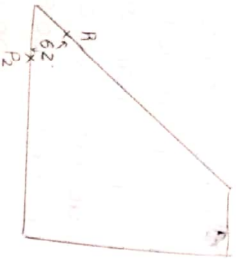
Diameter dimensioning - It should be used with dimensioning circles and arcs of more than 180°. Center marks are automatically drawn as you use the diameter dimensioning command.

- Select the dimension/Diameter command
- Command: dimdiameter

Radius dimensioning



Angular dimensioning



- Select arc or circle: Click on arc or circle P1
- Specify dimension line location: place dimension text location P2

⇒

Radius dimensioning - It should be used with/when dimensioning circle and arcs of more than 180° center marks are automatically drawn as you use the diameter/radius dimensioning command

- Select the diameter/radius command
- Command: dimradius
- Select arc or circle: Click on arc P1
- Specify dimension line location: Place dim text location P2

⇒

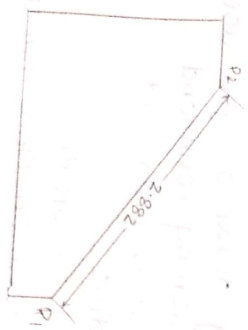
Angular dimensioning - It is used to create an angular dimension between two lines that form an angle. All that is necessary is the selection of the two lines and location of the dimension text. The degree symbol is automatically added

- Select the angular command
- Command: dimangular
- Select arc, circle, line <specify vertex>: click on first line that forms the angle P1
- Select the second line: click on second line that forms the angle P2
- Specify dimension arc line between and location place place dim text location

⇒

Aligned dimensioning - It aligns the dimension with angle of the object that you are dimensioning. The process is same as linear dimensioning.

Aligned Dimensioning



- Select dimensioning / Aligned pull down option.
- Command: dimaligned
- Specify first extension line origin: select P1
- Specify second extension line origin: select P2
- Specify dimension line location: place dimension text location.

Conclusion. Hence, we performed and drew different types of dimensioning.